Laser Cutter Safety Manual

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Important Policies

1. The Laser Cutter Safety Quiz is a quiz required first step to accessing the laser cutter at the Makerspace. You should read through the entire Laser Cutter Safety Manual before taking the Laser Cutter Safety Quiz.
2. **An in-person tutorial is required before using ANY laser cutter in the Makerspace.**
    a. If you have not attended an in-person tutorial, please ask an associate or officer to provide one to you. If you need a refresher, please ask an associate as well. All Makerspace associates are trained on how to use the laser cutter.

Introduction

The Makerspace offers one laser cutter available for select users. These machines are some of the most versatile in the Makerspace and can be used to cut and engrave wood, acrylic, aluminum, stainless steel, and more. This manual outlines some of the important policies and safety guidelines for these machines.

Laser Cutters

Machines

The Makerspace offers one laser cutter to all trained users **free of charge**: The Creality Falcon 2 Pro.

The Creality Falcon 2 Pro is a 40-watt CO_2 diode laser cutter used for both engraving and cutting. Non-metallic materials (wood, bamboo, cardboard, leather, ceramic, and acrylic) are suitable for both engraving and cutting. Stainless steel, aluminum, painted metal, and other materials are suitable for engraving only. This machine's build plate is 400 x 415 mm.

Note: You do not need to read a separate machine manual to pass the Laser Cutter Safety Quiz. However, before you use the laser cutter, it is recommended that you read the associated manual to learn the specifics of operating the machine.

Materials

Materials Offered at the Makerspace

The Makerspace supplies sheets of ⅛" Basswood that are free for all users. You may also bring in your own materials to cut, given that they are on the list of allowed materials. If you do, please ask an associate to help you.

Material Safety

Only certain materials are both allowed and physically able to be cut or engraved on the laser cutters. Some materials can be dangerous for both users and the machine itself, so this list is <u>not flexible</u>. **You must know what material you are using before you start a cut. Unknown materials can be just as or more dangerous than**

anything listed here. If you have a question about whether a material you own can be cut on the laser cutter, contact Dr. Jesse Smith, the MC3D advisor.
Unless otherwise noted below, **all materials must be ¼" or smaller** to be cut through, but almost any thickness of material can be engraved.

This list of materials is also posted above the Falcon2 Pro laser cutter and should always be considered before starting a cut.

Never Cut/Engrave

All of these materials can release noxious fumes or easily catch on fire when cut with a CO_2 laser. These fumes are dangerous to inhale and also harm the beam focusing lens. **Under no circumstances should these materials be cut or engraved on a laser cutter**.

1. **Unknown Plastics, or materials containing unknown plastics**
    a. This includes fake leather, polyester fabric, and fiberglass, which are not standardized in their material construction!
2. **Vinyl (or any compound containing vinyl), including PVC**
3. **Any fluorinated, chlorinated, or brominated compounds**
    a. Any materials containing fluorine, chlorine, and bromine cannot be cut.
4. **Polycarbonate**
5. **ABS**
    a. ABS is a common 3D printable plastic and is used in Lego bricks. This material emits harmful fumes when exposed to melting temperatures.
6. **Coated Carbon Fiber**
7. **Polystyrene Foam (Styrofoam) and Polypropylene foam**
8. **Any type of food**

Difficult to Cut/Engrave

These materials tend to be difficult to cut and engrave because they readily melt on contact with the laser. This can lead to discolored or gooey cuts that look low-quality.

1. HDPE/Milk Bottle Plastic
    a. This material is common in white kitchen cutting boards.
2. Epoxy
3. Other Polypropylene or Polystyrene Products (excluding foam)

Only Engrave

These materials are too strong to be cut through using the laser cutter. There are no dangers associated with trying to cut through these materials, but the raw power of the CO_2 laser is not enough to create a thick cut. Engraving these materials on high power and current settings works great!

1. Stone (including slate, marble, etc.)
    a. Any unknown stone materials should be verified before cutting.
    b. NO WET STONE
2. Ceramic Tile
3. Coated Metals/Anodized Aluminum
    a. You must know what the metal coating is made of to ensure that it is not vinyl-based or does not contain any other dangerous material. Most water bottles are fine. If engraving water bottles, ask an associate how to use the water bottle engraving accessory.

Safe to Cut/Engrave

These materials are safe to both cut and engrave on the Creality Falcon 2.

1. Most Wood
    a. Try to avoid oily or resinous woods. Any woods with a glue-based backing are considered to contain unknown plastic and thus cannot be used.
2. Plywood
3. Cardboard
4. MDF
5. Paper/Cardstock
    a. Use a very low power (~10%) and current (~15%) for an initial test! Using excessively high power or current can lead to burnt edges and fire.
6. Acrylic
7. Corkboard
8. Felt or Cotton Fabric
9. Leather/Suede (only cuttable up to ⅛")

General Use Policies

1. Before using the laser cutter, <u>you must read</u> the associated operating manual or watch the attached online tutorials to learn how to use it safely and properly.
2. After starting a job on any laser cutter, you **<u>must stay with the machine while it is running</u>**. Any time the laser is actively cutting or engraving, you **<u>are not allowed to walk away</u>** from the laser cutter.

The cutter will likely be running a job from your computer, so allot ample time to complete the print when you start a job

a. If you need to leave the room, you can pause the machine and resume it when you return.

3. In case of emergency, when using the Falcon2, press the **red E-Stop button** to immediately stop power flow into the machine. You may also flip the power switch or turn the key to cut power flow.

4. **Never, under any circumstances, operate a laser cutter with the top cover open**.

a. If any part of the machine is open while cutting, the laser beam can escape the enclosure, potentially causing immediate irreparable damage to the eyes and skin!

Safety Policies

There are some notable safety precautions that <u>must</u> be followed in order to ensure that the laser cutter will not cause any harm.

Eye Protection

REMEMBER: Always wear safety glasses as personal protective equipment (PPE) when in the vicinity of any lasers.

<u>Before You Start</u>

1. Be careful to keep fingers out of any moving parts, including the exhaust fans and motor rails.
2. Make sure you are using an approved material.
    a. If possible, try using an available piece of scrap material for your cut instead of using a new sheet!
    b. Always aim to use the smallest, least intrusive amount of material possible. This means keeping your designs as close to the edges of the material as possible.
3. Place your material in the laser cutter enclosure.
4. If using for the first time or the laser cutter has been inactive for a while, calibrate the laser and the camera.
5. Turn on the air ventilation system on the bottom shelf of the cart housing the laser cutter.
6. Verify that the red E-Stop button is rotated clockwise, and the power key is turned to the right.
7. **DO NOT USE THE LASER CUTTER WHEN:**

a. The top cover/lid is open
b. Any side panels are open
c. The exhaust fan is off
d. Any part of the machine looks broken
    - If this is the case, contact an associate!
e. There is a significant buildup of debris in the catch tray
    - This can cause a fire, so ask an associate if you are unsure.

While the Laser Cutter is Active

1. You <u>must</u> stay with your project at any time the laser cutter is active.
2. Monitor your cut to make sure nothing catches on fire.

After the Laser Cutter is Finished

1. Wait a short while (~30 seconds) for the fume extractor to clear all of the smoke in the laser cutter chamber.
    a. You may need to wait longer if lots of smoke was produced. Try to avoid opening the laser cutter cover if there is still smoke visible in the chamber.
2. Open the cover and take your piece off of the bed. If there is a usable amount of scrap left, place the remaining material in the scrap piles available.

IN CASE OF FIRE

Fires are the most common and most dangerous safety hazard when using a laser cutter. Small fires are far more common than large fires, but neither happen often (let's hope not at all). Some common causes of fire include:

1. Trying to cut through too thick of a material.
2. Trying to cut a disallowed material.

3. Using too high of a power/current setting for the material you are cutting.
4. Running the laser cutter when uncalibrated.

While it is unlikely for a fire to occur, it's important to understand the necessary steps to put out a fire in a laser cutter.

Small Fire

If the fire is **small** (controlled, small flames, confined to one part of the material), then do the following:

1. **Keep the lid closed**
2. Pause the job.
3. Wait for the fire to dissipate on its own. The exhaust fan system should keep oxygen away from the flame, and it should go out quickly.
4. If the fire does not go out, continue to the large fire prevention steps.
5. After the fire is out, **alert an associate** that a fire occurred.

Large Fire

If the fire is **large** (uncontrolled, large flames, consuming most of the material in the chamber, and possibly at risk of damaging the machine) then do the following:

1. Stop the cut immediately by pressing the red E-stop button.
2. Get the fire extinguisher located in the Sutton hallway near SSC 243.
    a. This is a specialty fire extinguisher that expels argon gas instead of normal liquid fire retardant, which can damage the sensitive electronics in the machines.
    b. Avoid using a fire extinguisher outside of the 3D printer/laser cutter room unless absolutely necessary.

c. Unplugging the laser cutter is unnecessary since the red E-stop button should have cut the power by this step. Make sure air is still ventilating within the machine. Once the fire is out, the argon gas should be filtered out.

3. Open the lid of the machine, and use the fire extinguisher to put out the fire.
4. After the fire is out, **alert an associate** that a fire occurred.

Once you have finished reading this manual, please take the

Laser Cutter Safety Quiz

Online Resources

Lightburn Crash Course
Beginning Laser Engraving
10 Project Ideas for Falcon2